Product Inventory Management System

Subject: IM102 CRUD Operations Project

Group Members: Member 1, Member 2, Member 3, Member 4

Instructor: Instructor Name

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1. Introduction

The Product Inventory Management System is a desktop application that helps users record and manage product information.

The system uses a graphical interface for CRUD operations and stores records in SQLite so data remains available after reopening the program.

2. Project Objectives

General Objective: To develop a simple inventory system that demonstrates complete CRUD operations using Python, Tkinter, and SQLite.

Specific Objectives: create a product entry form, save records, display records in a table, update selected records, delete selected records, calculate dashboard totals, and validate invalid input.

3. Functional Requirements

The system must add, view, update, and delete products. It must clear fields, refresh dashboard totals, and display pop-up messages for success, confirmation, validation, and database errors.

4. Tools and Technologies

Component	Tool	Purpose
Programming Language	Python	Application logic
GUI Library	Tkinter	Windows, forms, buttons, and pop-ups
Table Widget	ttk.Treeview	Display inventory records
Database	SQLite	Save product information
Development Tool	VS Code or IDLE	Write and run the program

5. Data Design

The database uses one table named products.

Column	Data Type	Constraint
product_id	INTEGER	Primary Key, Auto Increment
product_name	TEXT	Required
category	TEXT	Required
quantity	INTEGER	Nonnegative
unit_price	REAL	Nonnegative

6. File Design

File or Folder	Purpose
main.py	Contains the Tkinter interface and event handlers
database.py	Contains SQLite setup and CRUD functions
inventory.db	Stores product records and is generated automatically
documentation/flowchart.mmd	Contains the system flowchart
screenshots/	Stores screenshot evidence for testing

7. Flowchart and Algorithm

Process: Start -> Load Database -> Display Products and Totals -> Select Operation -> Validate or Confirm -> Save Changes -> Refresh Table and Totals -> Show Message.

Add Product Algorithm

1. Enter product details.
2. Validate required fields, quantity, and price.
3. Insert valid product into SQLite.
4. Refresh table and totals.
5. Show success message.

Read Product Algorithm

1. Open SQLite database.
2. Retrieve product records.
3. Display records in the table.
4. Calculate dashboard totals.

Update Product Algorithm

1. Select a product.
2. Edit product details.
3. Validate new values.
4. Update selected record in SQLite.
5. Refresh table and totals.

Delete Product Algorithm

1. Select a product.
2. Ask for confirmation.
3. Delete confirmed record.
4. Refresh table and totals.

5. Show success message.

8. Exception Handling Plan

Error Condition	Expected Pop-up Message
Empty product name	Please enter a product name.
Quantity contains letters	Quantity must be a whole number.
Negative quantity	Quantity cannot be negative.
Invalid price	Please enter a valid price.
Update without selecting	Please select a product to update.
Delete without selecting	Please select a product to delete.
Delete confirmation	Are you sure you want to delete this product?
Database error	Unable to save changes. Please try again.

9. Testing Evidence

Test ID	Test Condition	Expected Output	Actual Output	Remarks
TC01	Add a valid product	Product added and displayed	To be recorded	Pending
TC02	Enter letters as quantity	Error pop-up displayed	To be recorded	Pending
TC03	Enter negative stock	Input rejected	To be recorded	Pending
TC04	Update product stock	New quantity displayed	To be recorded	Pending
TC05	Delete selected product	Product removed after confirmation	To be recorded	Pending
TC06	Cancel deletion	Product remains unchanged	To be recorded	Pending
TC07	Close and reopen program	Saved changes remain	To be recorded	Pending
TC08	Add or delete product	Dashboard totals recalculate	To be recorded	Pending

10. Conclusion and Recommendations

The project demonstrates GUI development with Tkinter, CRUD implementation, SQLite persistence, input validation, and error handling.

Future improvements may include product search, low-stock alerts, inventory reports, authentication, CSV export, and printable reports.

11. Individual Contributions

Member	Contribution
Member 1	GUI design and Tkinter implementation

Member	Contribution
Member 2	Database design and CRUD functions
Member 3	Input validation, testing, and error handling
Member 4	Flowchart, screenshots, and documentation

12. Project Summary

The Product Inventory Management System is a simple desktop CRUD application for managing product records. It focuses on one product table, complete CRUD operations, validation, automatic dashboard totals, and organized documentation.